

5.5 Oscillations

This topic covers simple harmonic motion and damping.

This topic may be studied using applications that relate to, for example, the construction of buildings in earthquake zones.

This unit includes many opportunities for developing experimental skills and techniques by carrying out more than just the core practical experiments.

Candidates will be assessed on their ability to:

143	understand that the condition for simple harmonic motion is $F = -kx$, and hence understand how to identify situations in which simple harmonic motion will occur
144	be able to use the equations $a = -\omega^2 x$, $x = A \cos \omega t$, $v = -A\omega \sin \omega t$, $a = -A\omega^2 \cos \omega t$, and $T = \frac{1}{f} = \frac{2\pi}{\omega}$ and $\omega = 2\pi f$ as applied to a simple harmonic oscillator
145	be able to use equations for a simple harmonic oscillator $T = 2\pi \sqrt{\frac{m}{k}}$, and a simple pendulum $T = 2\pi \sqrt{\frac{l}{g}}$
146	be able to draw and interpret a displacement-time graph for an object oscillating and know that the gradient at a point gives the velocity at that point
147	be able to draw and interpret a velocity-time graph for an oscillating object and know that the gradient at a point gives the acceleration at that point
148	understand what is meant by <i>resonance</i>
149	CORE PRACTICAL 16: Determine the value of an unknown mass using the resonant frequencies of the oscillation of known masses
150	understand how to apply conservation of energy to damped and undamped oscillating systems
151	understand the distinction between <i>free</i> and <i>forced oscillations</i>
152	understand how the amplitude of a forced oscillation changes at and around the natural frequency of a system and know, qualitatively, how damping affects resonance
153	understand how damping and the plastic deformation of ductile materials reduce the amplitude of oscillation.